

HW#1

You can submit this HW as a teamwork of teams of a maximum of 2 students, but each individual must submit separately.

Consider a rocket with the following parameters:

Exhaust mass flow rate: $\dot{m}^0=8740 \text{ kg/s}$

Exhaust velocity: $u_e=2600 \text{ m/s}$

Exit pressure: $p_e=71 \text{ kPa}$

Exit temperature: $T_e=300 \text{ K}$

Initial gravity: $g_0=9.81 \text{ m/s}^2$

Ambient pressure:

If $h < 11 \text{ Km}$, changes with altitude according to:

$$p(h)=p_0(1-[L \times h]/T_0)^{5.256}$$

$$p_0=101325 \text{ Pa}$$

L =temperature lapse rate: $L=0.0065 \text{ K/m}$,

h : altitude in meters

$$T_0=288.15 \text{ K}$$

For $h > 11 \text{ Km}$, it changes:

$$p(h)= p_0 \times \exp(-g_0 \times M \times h / (R \times T_0))$$

$M = 0.029 \text{ kg/mol}$ (Molar mass of Earth's air)

Exit area: $A_e=11.35 \text{ m}^2$

The initial mass of the rocket: $M_0=1,720,000 \text{ kg}$

Burnout mass of the rocket: $M_b=680,000 \text{ kg}$

Gravitational force changes with altitude, given by

$$g(h)=g_0[R_e/(R_e+h)]^2,$$

where $g_0=9.81$, $R_e=6371 \text{ km}$ is the Earth's radius and h is the altitude in meters.

The rocket's angle of ascent θ varies with altitude, given by the relation

$$\text{Arctan}[v_0/(u_0+h)],$$

, where $v_0=200\text{m/s}$ and $u_0=800\text{ m/s}$, and h is the altitude in meters.

For simplicity, the drag force F_d is given by the formula:

$$F_d=C_d\cdot\rho(h)\cdot A_e\cdot(v(h)^2)/2$$

where $C_d=0.5$ is the drag coefficient, $\rho(h)$ is the air density at altitude h , and $v(h)$ is the velocity of the rocket at altitude h .

If $h < 11\text{ Km}$, density changes according to:

$$\rho(h)=\rho_0(1- [L\times h]/T_0)^{4.256}$$

$$\rho_0=1.225\text{ kg/m}^3$$

L , T_0 and h as described before.

If $h > 11\text{ Km}$, density changes according to:

$$\rho(h)= \rho_0\times\exp(-g_0\times M\times h/(R\times T_0))$$

$$\rho_0 = 1.225\text{ kg/m}^3$$

Other parameters as given in pressure formula

Using MATLAB (or any other software), do the following:

1. Compute $h_{\max}$ and h_b .
2. Plot ρ , p and g with altitude until $h_{\max}$.
3. Plot altitude-dependent thrust: $\tau(h)$ vs. h .
4. Plot the effective exhaust velocity at each altitude: $u_{eq}(h)$ vs. h .
5. Plot the altitude-dependent specific impulse: $I_{sp}(h)$ vs h .
6. Compute burn time.
7. Compute and plot a_0 , a_{b-} and a_{b+} .
8. The velocity rise at any altitude, $v(h)$, accounting for drag and gravity changes.

Clearly write all your equations. Submit a report for this HW with the figures and data are put inside the report and the code put at the end of the report.

Hint-1:

Since the drag force is a function of velocity, and the velocity is influenced by the drag force (in velocity rise formula), you need to solve this iteratively.

Steps to account for this interdependence:

1. **Initial Guess for Velocity:** Start with an initial guess for velocity v_0 .
2. **Compute Drag Force:** Using the current guess for velocity, calculate the drag force.
3. **Update the Velocity:** Compute the acceleration using the drag force and gravitational force, then update the velocity.
4. **Iterate:** Repeat the process, updating velocity and drag force at each step, until convergence is achieved (i.e., the change in velocity between iterations is small enough).

Hint-2:

h can be computed from the below formula

% Compute altitude (h) using the provided formula

```
argument = 1 - (1 - 1 / R) * time(i) / burn_time;
```

```
argument = max(argument, eps); % Ensure it doesn't go below machine epsilon
```

```
h = u_eq(i) * burn_time * R / (R - 1) * ...
```

```
((argument) * log(argument) - (argument) + 1) - (1 / 2) * g_e * time(i)^2;
```

Hint-3:

Integrals of the second and third terms of the velocity rise equation can be performed numerically:

% Compute the two integrals numerically: %

```
1. Integral for gravitational force term gravitational_integral = trapz(time, g_e .* cos(atan(v0 ./ (u0 + altitude)))) .* time);
```

```
2. Integral for drag force term drag_integral = trapz(time, drag_force ./ mass);
```

Final velocity rise equation:

```
velocity_rise = u_eq .* log(M0 ./ mass) - gravitational_integral - drag_integral;
```